Geospatial Analysis of Health Equity in Massachusetts

Using CDC/ATSDR Social Vulnerability Index (2022)

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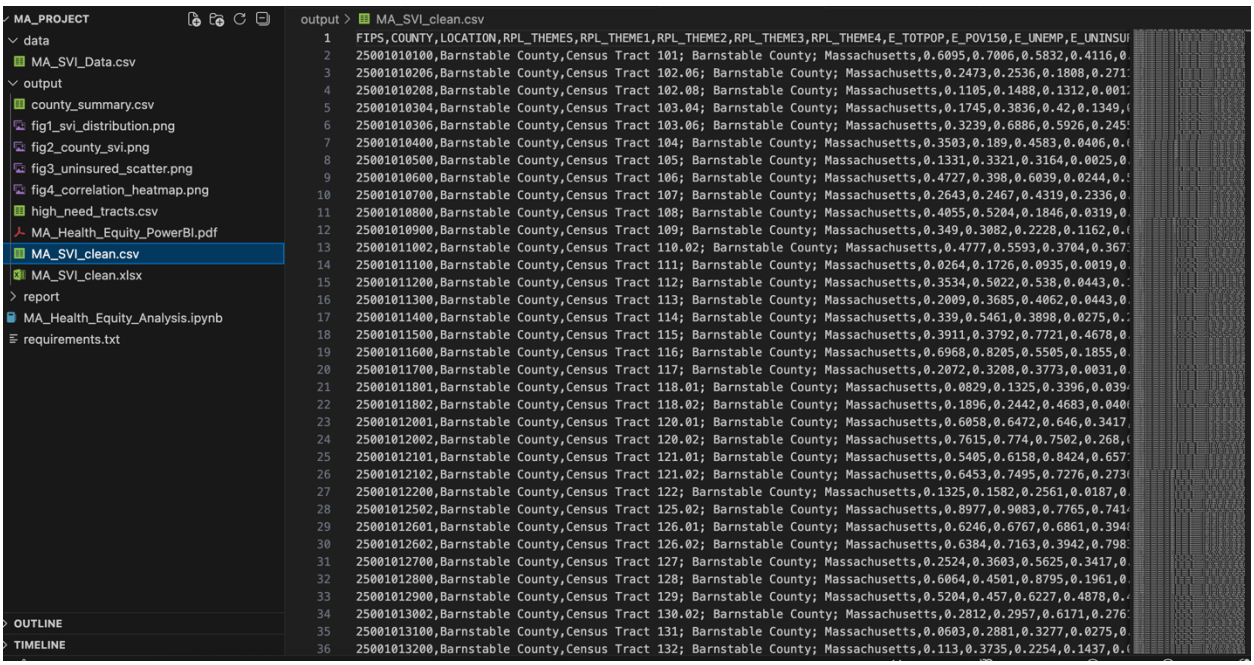
1. Executive Summary

The impending 2026 shifts in federal subsidies and tax credits will fundamentally alter the health insurance landscape, placing certain Massachusetts populations at a heightened risk of losing coverage. This project utilizes public Social Vulnerability Index (SVI) datasets to develop a data-driven framework that identifies these vulnerable groups. By engineering a custom "Subsidy Risk Score," this analysis translates complex socio-economic variables into actionable geospatial insights, equipping public policy makers with the tools needed to direct outreach efforts and safeguard coverage for the Commonwealth's most at-risk residents.



2. Methodology & Data Engineering

To ensure a targeted analysis, demographic and socioeconomic data for Massachusetts census tracts was preprocessed using Python (Pandas and NumPy). Missing values were systematically cleaned to prevent algorithmic bias.

Rather than relying on generalized vulnerability metrics, a specialized **Subsidy Risk Score** was engineered to directly reflect the barriers associated with subsidy dependence and navigation difficulty. Key variables were normalized to a 0-1 scale and weighted based on their impact on health insurance continuity:

- **Poverty Levels (EP_POV150 - 45% Weight):** Represents the primary driver of subsidy eligibility.
- **Uninsurance Rates (EP_UNINSUR - 35% Weight):** Highlights populations already struggling with coverage continuity.

- **Limited English Proficiency (EP_LIMENG - 20% Weight):** Identifies groups facing significant navigational and administrative hurdles when interpreting complex policy changes.

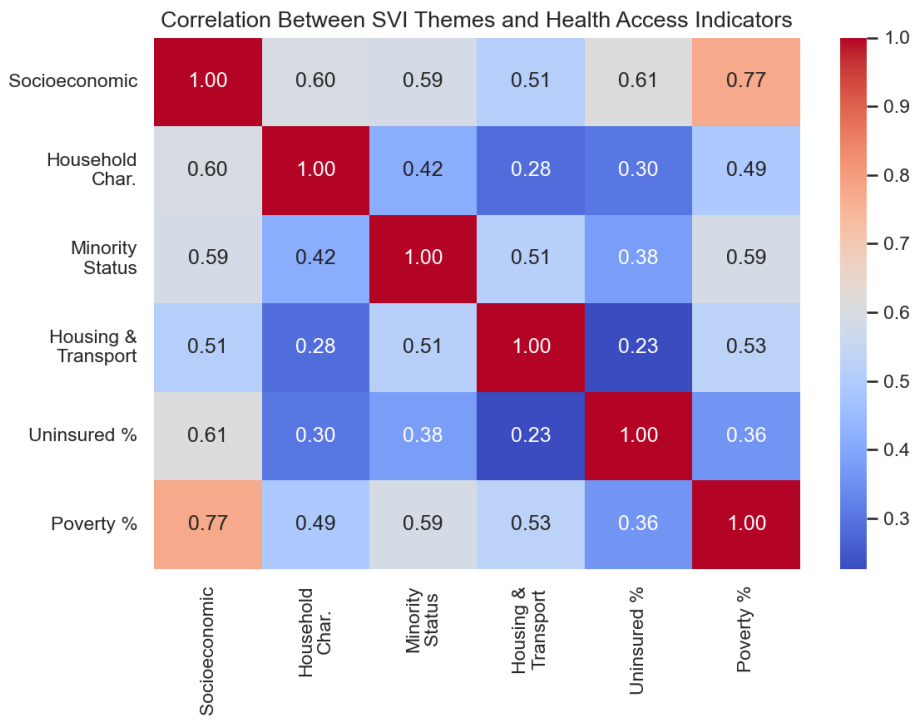


Figure 1: Correlation matrix confirming socioeconomic vulnerability (r=0.77 with poverty, r=0.61 with uninsured rate) as the dominant SVI theme.

This composite score bridges the gap between raw data and public policy needs, allowing for a precise risk ranking of communities. County-level summaries were computed as population-weighted averages to ensure that densely populated tracts contribute proportionally to the aggregate vulnerability score.

3. Geospatial Findings

The processed data was visualized using Power BI to identify high-risk zones across the state. The interactive dashboard reveals significant regional disparities:

- **Highest Vulnerability Counties:** Suffolk County emerged with the highest average overall SVI (0.675), alongside a staggering poverty rate of 25.03% and an uninsurance rate of 3.49%. Hampden County followed closely with an SVI of 0.639 and nearly 24% poverty. These regions represent the highest density of residents at risk of coverage loss.

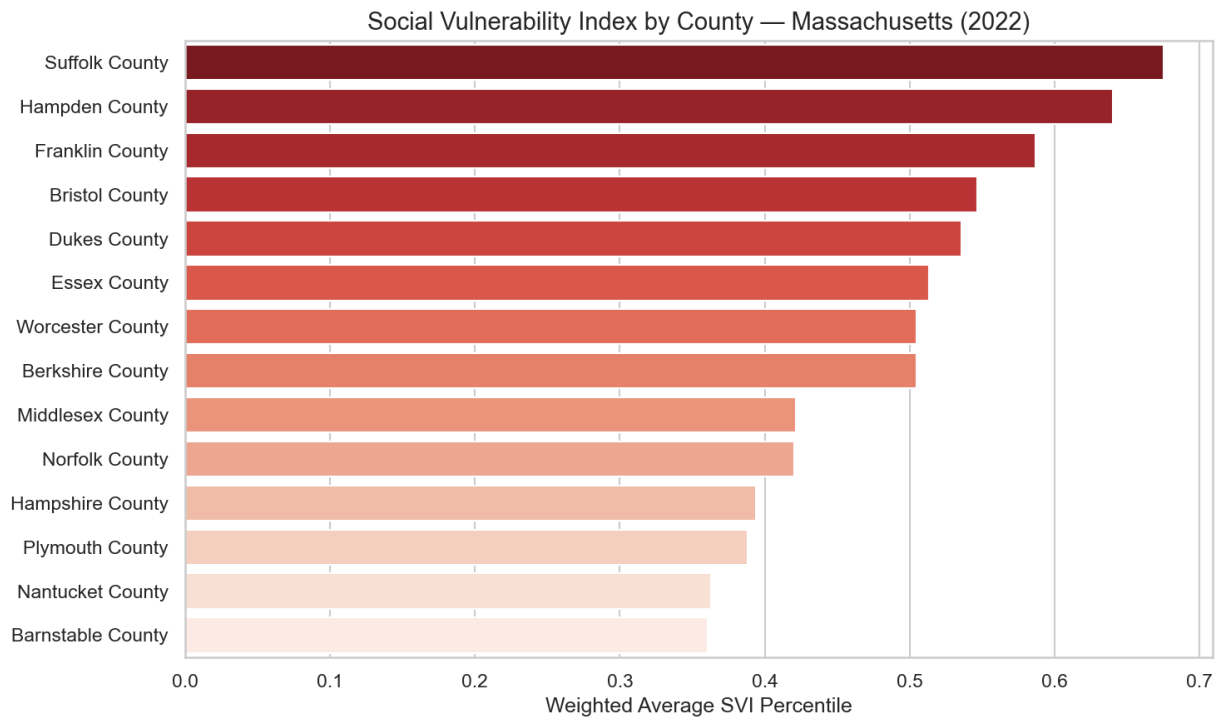


Figure 2: Population-weighted average SVI by county. Suffolk and Hampden counties lead in overall vulnerability.

- **Correlations:** The scatter plot analysis visually confirms a direct, positive correlation between socioeconomic vulnerability (Theme 1, SVI) and uninsurance rates. This proves that financial instability heavily dictates coverage fragility across all analyzed tracts.

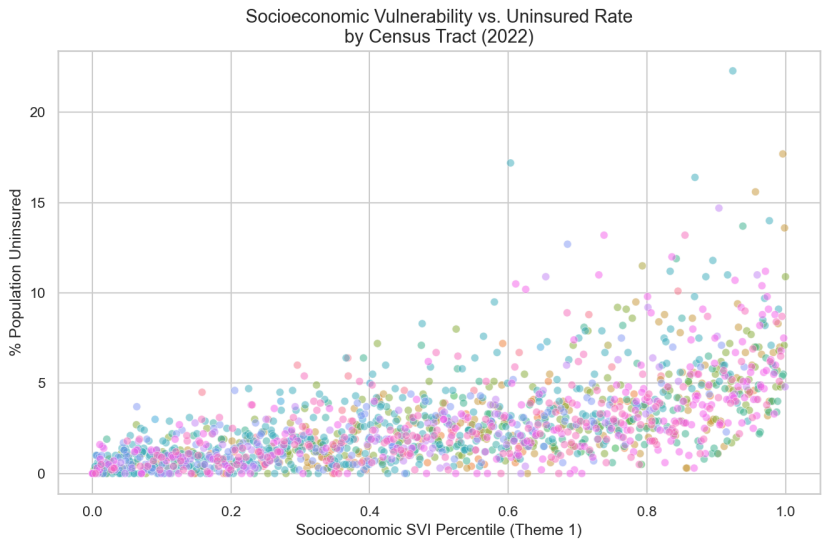


Figure 3: Positive correlation between socioeconomic SVI percentile and uninsured rate across 1,594 Massachusetts census tracts.

- **Hidden Pockets of Risk:** Interestingly, Dukes County presented the highest uninsurance rate (6.45%) despite a moderate overall SVI (0.535), indicating a unique regional challenge that requires specialized intervention.

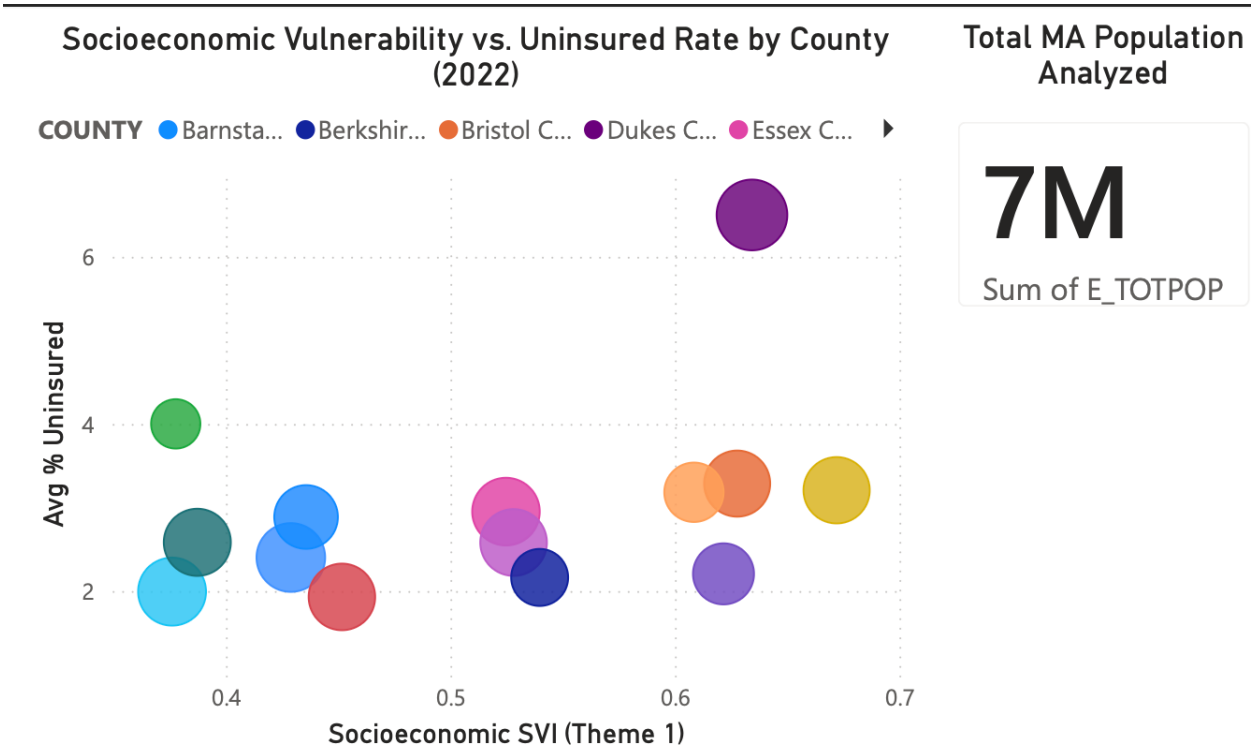


Figure 4: Interactive Power BI dashboard view highlighting county-level uninsured rate vs. socioeconomic vulnerability, with Dukes County (purple) as a notable outlier.

Across Massachusetts, 223 census tracts — home to approximately 955,984 residents — were identified as high-need zones, defined by an overall SVI ≥ 0.75 and uninsured rates in the top 25th percentile statewide. Suffolk County accounts for 53 of these tracts (231,703 residents), followed by Middlesex (42 tracts) and Essex (34 tracts).

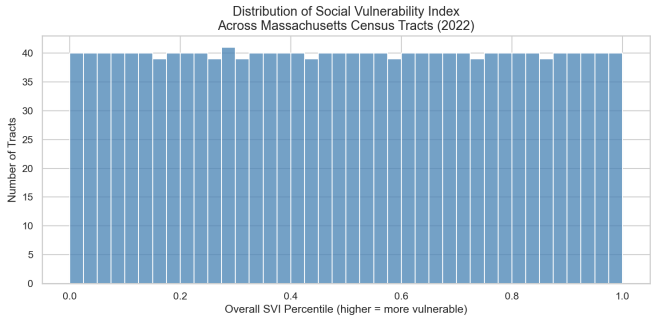


Figure 5: Distribution of SVI percentiles across MA census tracts, showing vulnerability is present across the full spectrum - not concentrated in a single region.

4. Policy Recommendations

To proactively combat coverage loss resulting from the 2026 federal subsidy changes, the following strategies are recommended for the Health Connector:

1. **Hyper-Local Outreach:** Deploy targeted community outreach programs specifically within Suffolk and Hampden counties, prioritizing census tracts with the highest Subsidy Risk Scores.
2. **Navigational Support:** For areas demonstrating high Limited English Proficiency (EP_LIMENG) alongside high poverty, heavily prioritize the distribution of translated materials and specialized, bilingual Navigators to assist with re-enrollment and subsidy recalculations.
3. **Investigate Outliers:** Conduct a secondary investigation into Dukes County to understand the root cause of its disproportionately high uninsurance rate, tailoring a unique intervention plan for the island's demographics.

5. Future Scope: AI and Diagnostic Interpretability

While this model successfully highlights geographic vulnerabilities, the next iteration of this analysis could involve predictive machine learning to forecast exact coverage-loss volumes. However, as public policy increasingly relies on complex models, transparency becomes paramount.

Future expansions of this project will integrate AI interpretability frameworks, similar to those utilized in analyzing medical imaging models (e.g., examining Vision Attention Maps for CNNs). Applying this level of rigorous evaluation to socio-economic prediction models ensures that the algorithms determining health equity gaps remain transparent, interpretable, and deeply trusted by public health clinicians and policy directors alike.